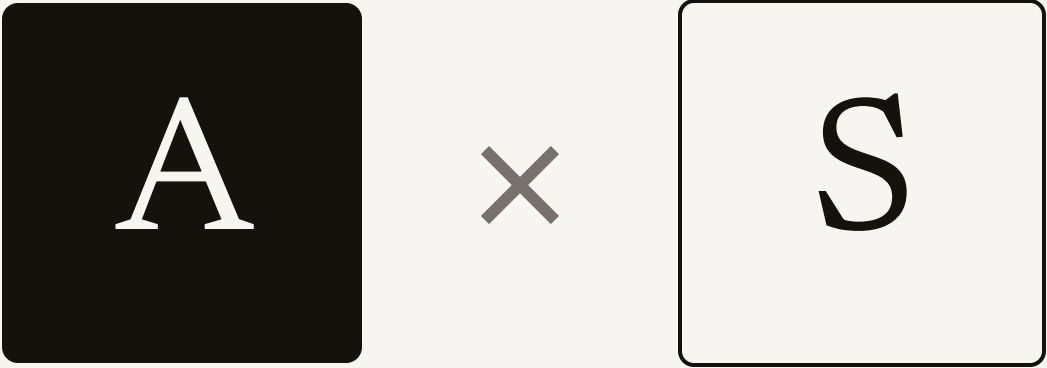




Apple vs. Samsung

A Strategic Economic Analysis



Apple won in court. *Samsung won the market.*

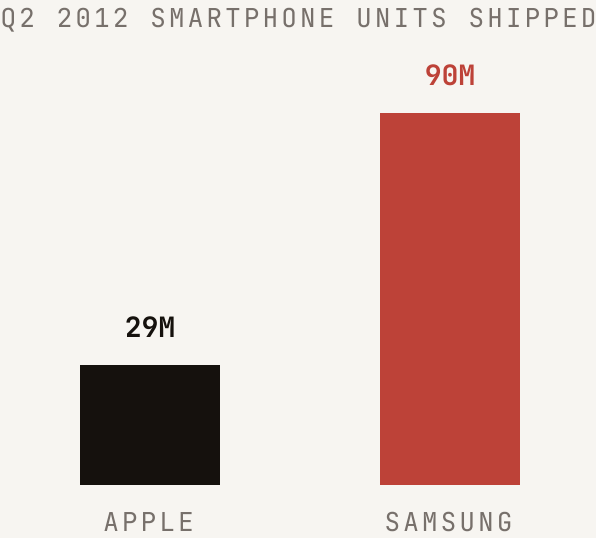
How can both be true?



AUGUST 2012 – VERDICT

\$1.05B

LARGEST PATENT VERDICT IN U.S. HISTORY



FROM QUESTION TO VERDICT

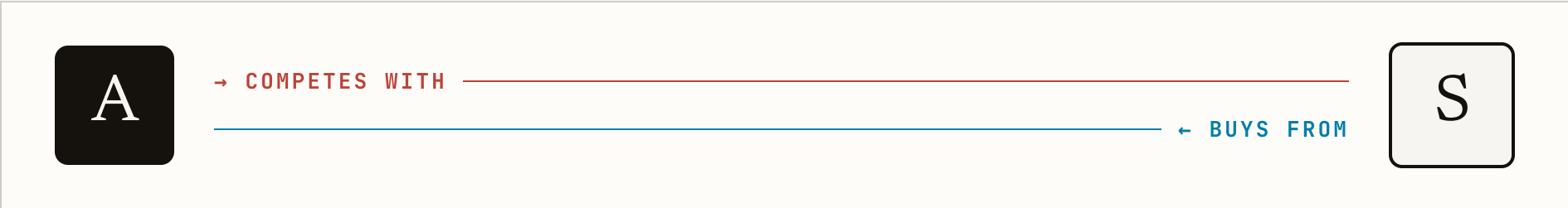
Our argument in *six beats*.

I	The Setup Why they couldn't just walk away.	SLIDE 5
II	Why Apple Sued Bargaining failure, three pressures, expert critique.	SLIDES 7-12
III	The Pivot Why the threat had a ceiling.	SLIDE 13
IV	The Logic Two parallel games at once.	SLIDES 14-15
V	The Evidence What investors actually saw.	SLIDES 16-17
VI	The Verdict Who won — and what it means.	SLIDE 18

I · ONE STRANGE RELATIONSHIP

Rivals *and* supplier–customer at the same time.

This duality — **coopetition** — constrains every move that follows in the dispute.



FACT	NUMBER
Samsung's share of iPhone manufacturing cost	26%
Apple's component purchases from Samsung (2010)	\$6B
Apple's share of Samsung's total revenue	4%

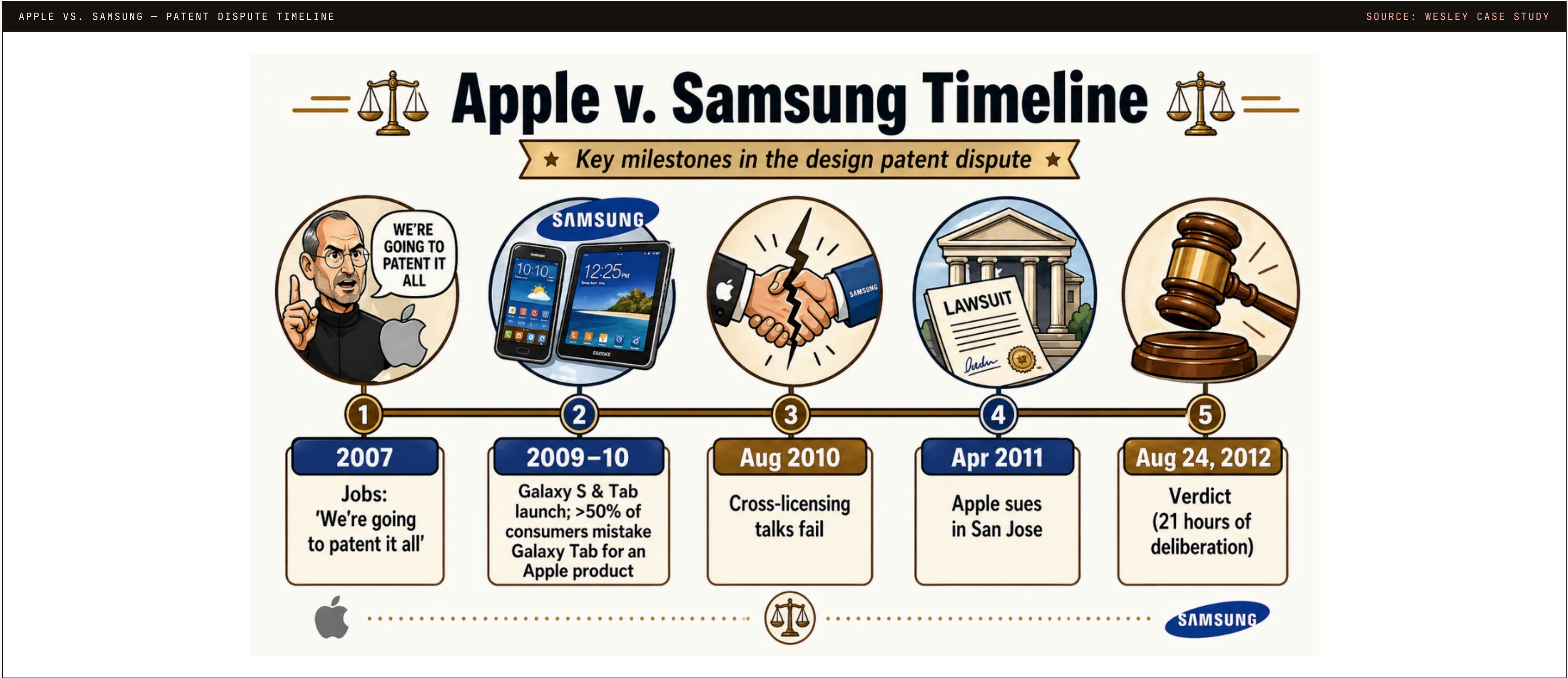
"Apple was reluctant to take legal action against Samsung, given its key role in supplying chips and components."

— WESLEY CASE STUDY

SEVEN YEARS IN FIVE BEATS

From Jobs's promise to the largest patent verdict in U.S. history.

2007 → 2012
5 BEATS • 1 VERDICT



BEFORE THE THREE PRESSURES • THE AUG 2010 COLLAPSE

Five reasons the deal couldn't close.

01

A ONE-WAY TAX, NOT A TRADE

Jobs offered Samsung an Apple license at **~\$30/phone + ~\$40/tablet** ($\approx$ \$250M/year per the trial record). It was a license fee — Apple did not propose to pay for Samsung's IP in return. *A one-way license is not a cross-license.*

02

ASYMMETRIC PORTFOLIOS

Apple's strong patents were **UI & design** — premium-priced. Samsung's strong patents were **3G wireless SEPs**, capped by FRAND duty. Different ammunition; structurally unequal value in any trade.

03

DESIGN PATENTS DON'T CROSS-LICENSE

Utility patents license a *function*; both sides can still differentiate. Design patents license *identity itself*. Letting Samsung use the iPhone's look would **destroy the differentiation premium Apple was suing to protect** (Hotelling, slide 10).

04

CREDIBILITY AGAINST OTHER DEFENDANTS

Apple was simultaneously suing **HTC, Motorola, Nokia, Amazon, Google**. Settling cheaply with Samsung would have collapsed every other threat. Refusing to license *was* the maximum-credibility move (slide 13).

05

HOLDOUT UNDER INFORMATION ASYMMETRY

Each side believed its own portfolio stronger. Without an agreed valuation, no Coase-style bargain converges — the *classic cross-licensing failure mode*.

The standard fix doesn't work when the patents protect identity, not function.

— WHY SUING WAS APPLE'S ONLY LEVER LEFT

II • APPLE DID NOT SUE OUT OF ANGER

Three economic forces pushed Apple *into court*.

Each one is the subject of one of the next three slides.

1 PRESSURE #1

Patent Hold-Up

Too many patents, no clean way out — and the standard fix (cross-licensing) had failed.

2 PRESSURE #2

Lost Differentiation

Samsung looked too much like Apple, and Apple's premium was eroding.

3 PRESSURE #3

Platform Tipping

Android was running away with the smartphone OS market — second place might not survive.

IF YOU REMEMBER NOTHING ELSE FROM THIS DECK — REMEMBER THESE THREE.

WHY APPLE SUED — I

When no one owns the road, everyone pays a toll.

~10K patents needed
TO SHIP ONE SMARTPHONE

\$10B / year, industry-wide
ON PATENT LITIGATION BY 2011

Apple & Google each spent *more on lawsuits than R&D*.
— ECEB SECTION 3.1

Cross-licensing is the textbook fix. It failed in Aug 2010.

PATENT STRATEGY — COMPARED

	APPLE	SAMSUNG
Offensive	Sue Samsung for copying iPhone	Counter-sue over wireless patents
Defensive	"Patent everything" — wall of patents	Build own patent stockpile
Bargaining chip	Win cases → better licensing terms	Wireless patents as leverage

WHY APPLE SUED - II

Bertrand Competition & Hotelling, in plain sight.

BERTRAND COMPETITION

Identical products → price war → zero profit.

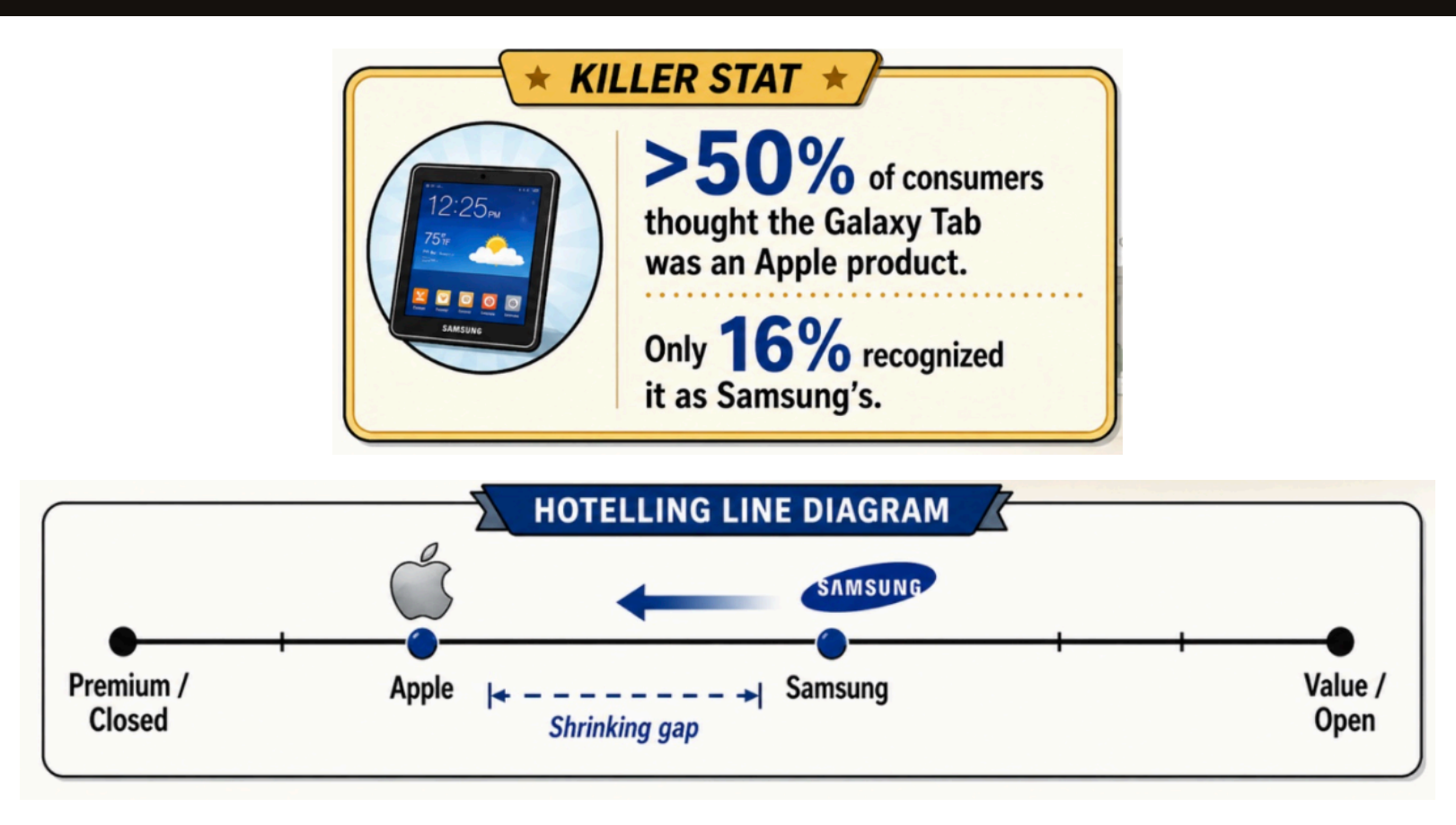
HOTELLING

The closer rivals stand on the "taste line," the harder they compete.

IMPLICATION

Samsung's copying moved it toward Apple → eroded the premium.

FIG. 2 • KILLER STAT + HOTELLING LINE



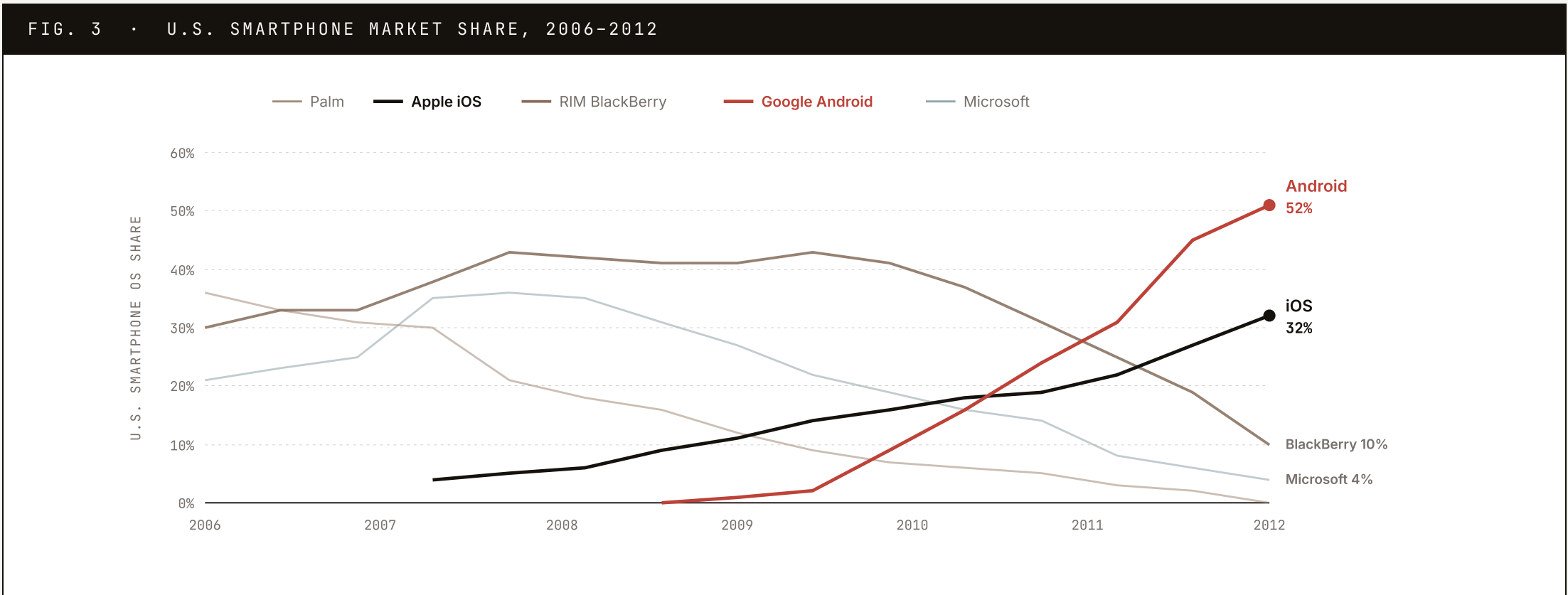
CAPTION: AS SAMSUNG'S POSITIONING APPROACHED APPLE'S, THE DIFFERENTIATION PREMIUM ERODED.

WHY APPLE SUED – III

Network effects don't reward second place.

TWO PLATFORMS, TWO GAMES

	IOS (APPLE)	ANDROID (SAMSUNG+)
Standard	Closed	Open
Network effect	Apps → users	Same — across many OEMs
Switching costs	High (apps, iCloud)	Low between brands
Game	Competes FOR the market	Competes IN the market



SOURCE: CASE EXHIBIT 10 – CANALYS, SILICON VALLEY INSIDER, COMSCORE, GARTNER.

THE VIEW FROM THE BENCH

"Patents are part of the problem."

*"They are not an ideal solution to the problem [of creating adequate incentives for invention]; instead they are **part of the problem**. Although they are supposed to be limited to inventions that are novel, useful, and non-obvious, they are granted rather **promiscuously** by the Patent and Trademark Office. And while they can be challenged in court, **jurors tend to be biased in favor of patent holders**... Patent **'trolls'**... purchase large numbers of patents in the hope of using the threat of a patent-infringement suit to extort a patent-license fee from a company that makes a similar product..."*

RICHARD A. POSNER · SLATE, OCT 15, 2012 · CASE P. 2

POSNER'S CLAIM → EVIDENCE FROM THIS CASE

1. PTO grants are promiscuous.
'381 was 62 pages with 20 broad claims; the USPTO later issued a **non-final rejection** of the same patent in 2013.

2. Jurors favor patent holders.
Verdict in **21 hours**; jury upheld every Apple claim despite documented prior art (LG Prada, Knight-Ridder tablet, MERL touch demos).

3. Patents become weapons, not incentives.
\$10B/yr industry-wide on patent litigation by 2011; Apple and Google each **spent more on lawsuits than R&D** (case p. 2).

4. Patents are part of the problem.
Defensive arms race; ~10K patents per smartphone (slide 09) — the system *raises* the cost of inventing in dense fields.

*The aftermath proved Posner right: **AIA inter partes review** (2011) · **Alice v. CLS Bank** (2014) · **TC Heartland** (2017) · **SCOTUS 8-0** in this very case (2016).*

III · THE PIVOT – APPLE'S THREAT HAD A CEILING

"I will spend my last dying breath... every penny of Apple's \$40 billion in the bank... I'm going to destroy Android."

– STEVE JOBS

STRATEGIC COMMITMENT – 3 CRITERIA

Observable?

Public lawsuit + injunctions sought

✓ YES

Understandable?

Copy us → face \$1.05B+ costs

✓ YES

Credible?

Sunk costs, reputation on the line

△ PARTIAL

WHY ONLY PARTIAL?

PUSH TO SUE

Already suing **HTC, Motorola, Nokia, Amazon** and **Google** (Exhibit 2) → softening on Samsung breaks every other threat.

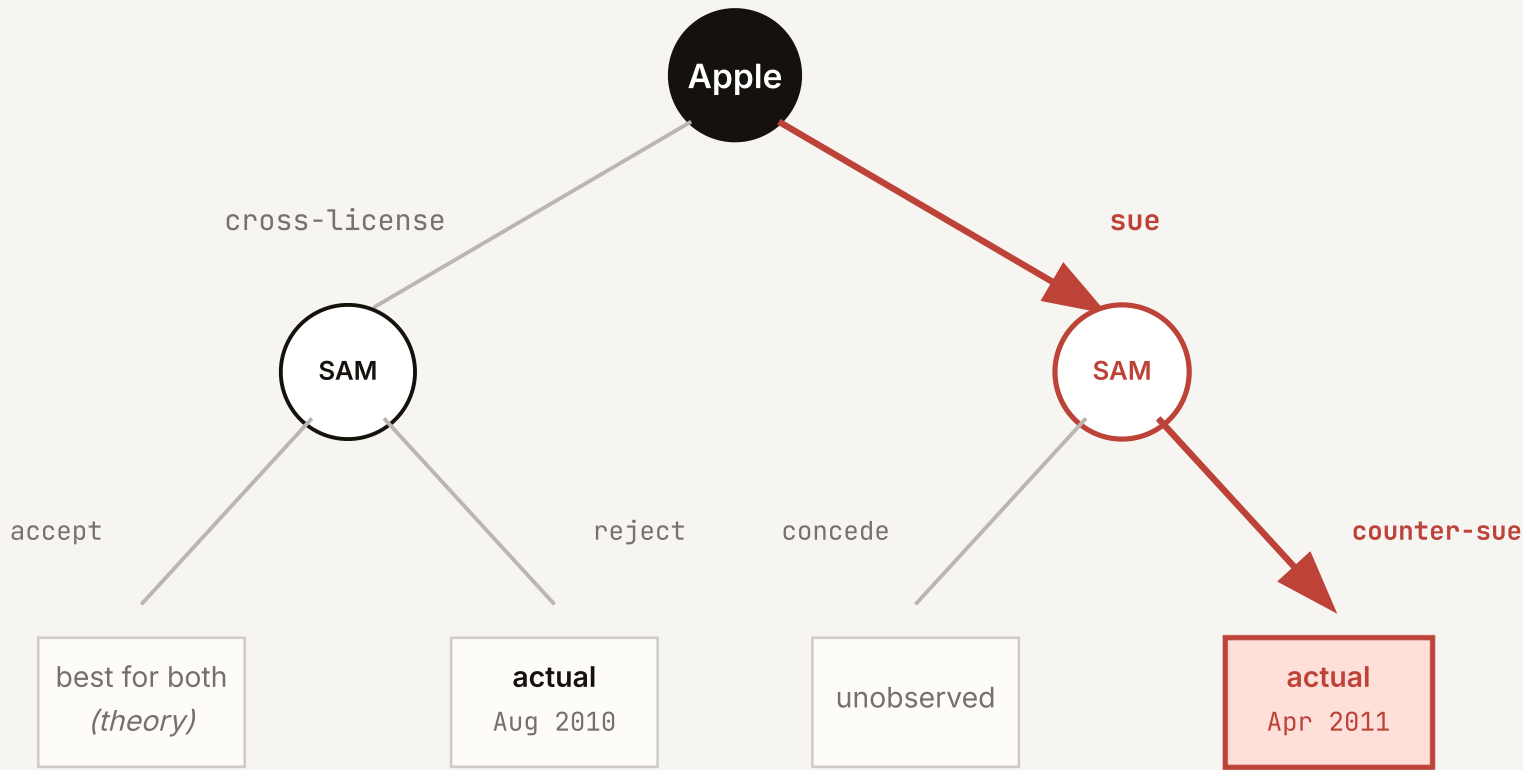
trigger strategy / reputation in a repeated game · suit is the minimum-viable move

PULL TO HOLD BACK

Samsung supplies Apple's chips, screens, memory → can't actually push to ban.

coopetition / hold-up

GAME TREE – HIGHLIGHTED: ACTUAL PATH



CROSS-LICENSE → REJECT → SUE → COUNTER-SUE

APPLE'S THREAT LACKED FULL CREDIBILITY – BOTH SIDES SETTLED IN COURT.

IV · COOPETITION LOGIC

Apple and Samsung play *two games at once* – and the second one saved the relationship.

GAME 1 Smartphone Competition

PRISONER'S DILEMMA

	SAM: DIFFERENTIATE	SAM: IMITATE
APPLE: TOLERATE	Mod π , Mod π	Low π , High π
APPLE: SUE	–Legal, 0	–Legal, –Legal

GAME 2 Supply Chain Cooperation

REPEATED GAME

	SAM: RELIABLE SUPPLY	SAM: DISRUPT
APPLE: CONTINUE	★ \$6B+, \$6B+ NE	Crippled, Short gain
APPLE: DIVERSIFY	Transition cost, Lose customer	Both lose, Both lose

COOPERATION INEQUALITY

$$r \leq \frac{\pi^C - \pi^{NC}}{\pi^D - \pi^C}$$

r

DISCOUNT RATE

your impatience · ~5–10%

π^C

COOPERATE

\$6B / \$6B · supply & buy continues

π^{NC}

NO COOPERATION

price war wipes profit · **≈ \$0**

π^D

DEFECT ONCE

short-term spike, then relationship collapses

Long-term reward ÷ one-time temptation. With **\$6B/yr** at stake, the inequality holds easily.

WHO INNOVATES, AND WHY

Two effects, two players, two strategies.

REPLACEMENT EFFECT

Apple — the entrant.

Nothing to lose.

Pre-iPhone, Apple had \$0 smartphone revenue.

Big disruptive bet.

The touchscreen iPhone — bet the company on a new category.

EFFICIENCY EFFECT

Samsung — the incumbent.

Now has profits to protect.

Incremental gains: better screens, chips, scale.

R&D climbs incrementally.

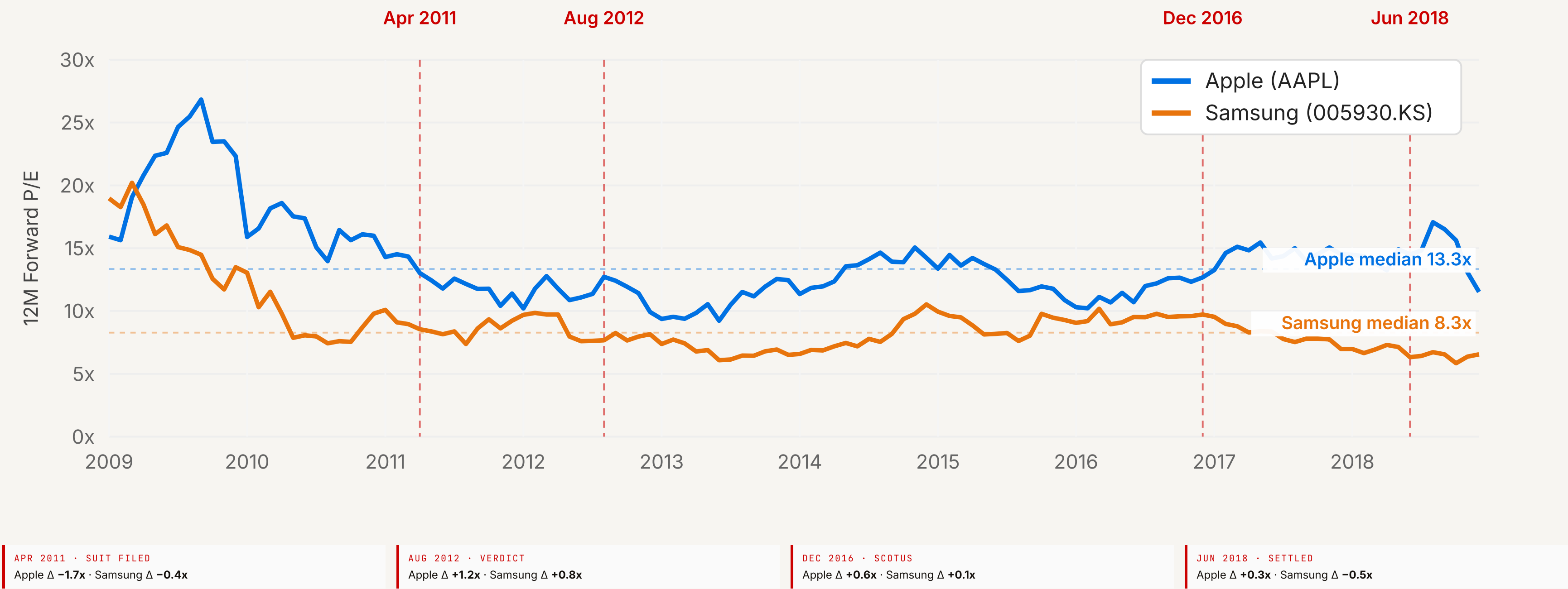
\$8.3B → \$9.1B (2010 → 2011).

THE PRINCIPLE

*"The more you have to lose,
the smaller your bets."*

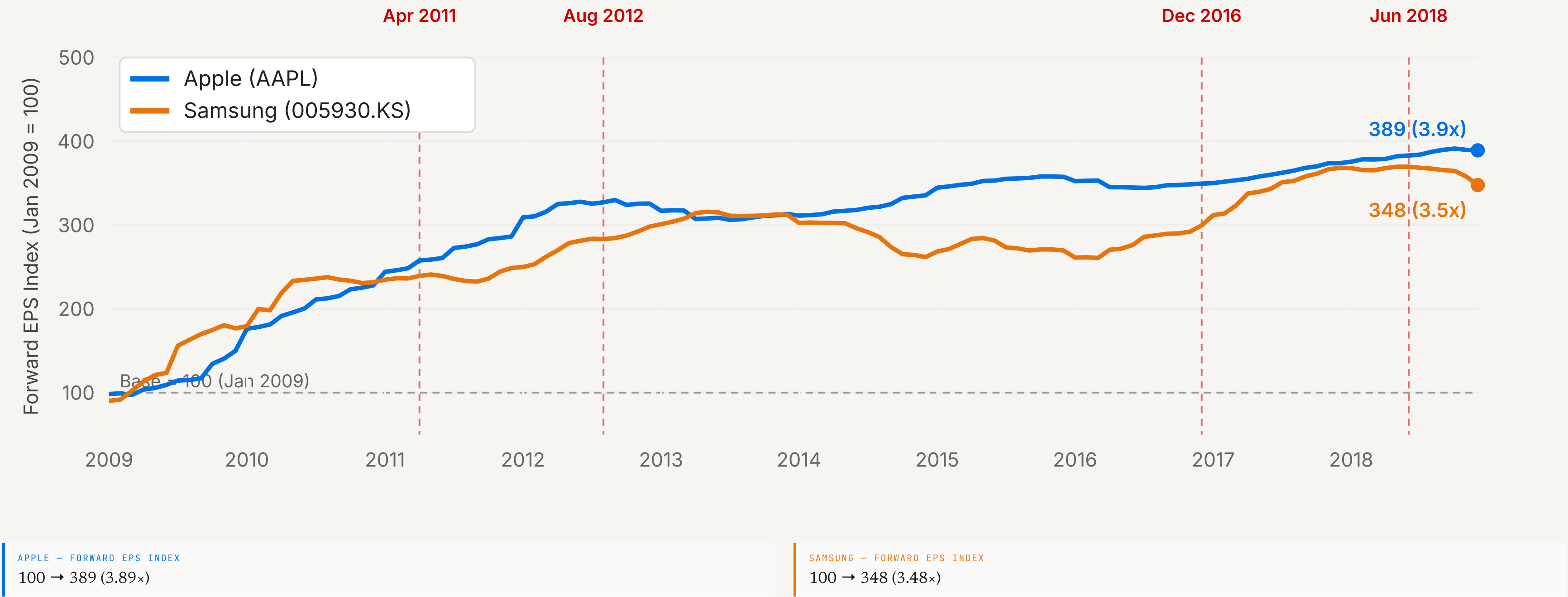
V • WHAT MARKETS SAID

Investors never re-rated either firm on lawsuit news.



WHAT EARNINGS DID

Earnings nearly 4× for both firms — through the entire dispute.



VI • THE VERDICT, FIVE YEARS LATER

Two scorecards. Two stories.

DAMAGES AWARDED

\$539M

8.4% OF SAMSUNG'S 2011 NET INCOME

GOAL OUTCOMES – APPLE VS. SAMSUNG

Apple

Goal	Outcome
 Damages	\$539M (reduced from \$1.05B) – <i>Partial</i>
 Ban Samsung products	Failed
 Slow Android	Failed (Android hit 52%)

SAMSUNG

Goal	Outcome
 Become #1 smartphone maker	✓ 90M vs 29M, Q2 2012
 Maintain supply relationship	✓
 Establish Android	✓

CLOSING THOUGHT 

“

When companies depend on each other,
patents are bargaining chips, not weapons.

”

Courts move slowly. Markets don’t wait.

EPILOGUE I · SAMSUNG'S SCALE

Six years of growth, in USD billions.

METRIC	2006	2007	2008	2009	2010	2011
Sales	89.4	106.0	110.1	106.8	133.8	148.9
R&D	*	*	*	5.8	7.9	9.0
Net Income	8.3	8.0	5.0	7.6	14.0	12.4
Net Margin	9.3%	7.5%	4.6%	7.2%	10.4%	8.3%
Equity	50.2	60.2	57.1	57.2	77.3	91.9

SOURCE: SAMSUNG AUDIT REPORTS. CONVERTED TO USD AT FRB H.10 PERIOD-AVERAGE KRW/USD: 2006 955.5 · 2007 929.3 · 2008 1,102.1 · 2009 1,276.9 · 2010 1,156.1 · 2011 1,108.3. *R&D BUNDLED IN SG&A PRE-2009 (K-GAAP).

EPILOGUE II • FIVE YEARS LATER

Seven years of litigation, quietly settled.

2016

U.S. SUPREME COURT — 8-0

Design-patent damages should be calculated only on the *copied component* (e.g., the front face), not the whole phone — narrowing the \$399M design-patent portion.

2018

RETRIAL & SETTLEMENT

\$539M awarded. A confidential settlement in June 2018 ended seven years of litigation.

& yet

THE QUIET TRUTH

Despite the war, the supply partnership continued.

CASE QUESTION I · MARKET DEVELOPMENT

How this case rewrote the rules of mobile.

01

DUOPOLY CEMENTED

By Q2 2012: **Android 52%** · **iOS 32%**. BlackBerry collapsed 43→4%. Litigation cost-of-entry left no room for a third platform.

02

DESIGN-AROUND CULTURE

Every Android OEM built legal-and-design teams to engineer around Apple's claims. Differentiation became *defensive engineering*, not aesthetic instinct.

03

TABLET COPYING CHILLED

Galaxy Tab 10.1 was redesigned for several markets; Samsung's industrial-design language pulled deliberately away from Apple's after 2012.

04

DAMAGES DOCTRINE NARROWED — SCOTUS 8-0 (2016)

Design-patent damages limited to **the copied component**, not the whole device — a rule change *directly traceable to this case*.

05

FRAND / SEP NORMS TIGHTENED

Samsung's wireless-SEP counter-suit clarified **FRAND** licensing duty — standards-essential patents cannot be priced like premium IP.

Apple won the case. Samsung won the market. The industry rewrote its rulebook.

— Q1 TAKEAWAY

CASE QUESTION II • THE COUNTER AT TRIAL

Strong on paper. Weak in the jury box.

UTILITY PATENT

U.S. #7,469,381 *"bounce-back"*

List scrolling and document translation, scaling, and rotation on a touch-screen display. 62 pages · 20 claims.

SAMSUNG'S GROUNDS (PER JUSTICE KOH)

1 · Inadequate written description	weak
2 · Anticipation by prior art	primary attack
3 · Statutory bars	not pressed
4 · Obviousness	primary attack

PRIOR ART CITED

DiamondTouch (MERL); Tablecloth (LaunchTile); academic touch-UI demos pre-2007 — all argued to show the elastic-feedback gesture before Apple's filing date.

OUTCOME

Jury upheld validity (2012). **The USPTO issued a non-final rejection of '381 in 2013** on prior-art grounds — Samsung's theory had merit it could not deliver to a jury.

Why Samsung lost despite real evidence: *clear-and-convincing* burden of proof + the *ordinary-observer* test rewards consumer confusion + juror bias toward patent holders (see slide 12 — Posner).

DESIGN PATENTS

D'677 · D'087 · D'889

D'677 black face · D'087 rounded rectangle · D'889 tablet form. *Each design patent has a single claim — the drawings themselves.*

SAMSUNG'S GROUNDS

1 · Anticipation by single prior art	primary attack
2 · Obvious to ordinary designer	primary attack
3 · Functional, not ornamental	narrow
4 · Non-infringement (look distinct)	parallel arg.

PRIOR ART CITED

LG Prada (2006); Sony "Jony" tablet sketches by Apple-internal designer Nishibori; the 2001 Knight-Ridder tablet concept video for D'889.

OUTCOME

Validity upheld. The *"ordinary observer"* test favored Apple given >50% Galaxy-Tab/iPad consumer-confusion data.

